第 8 章转录公式语法校验（非教材排版定稿）

pdf-215

$$Ax = b, \tag{8.1.1}$$

A

A

A

n

$n \geqslant 10^4$

A

$$\begin{cases} 8x_1 - 3x_2 + 2x_3 = 20, \\ 4x_1 + 11x_2 - x_3 = 33, \\ 6x_1 + 3x_2 + 12x_3 = 36, \end{cases} \tag{8.1.2}$$

$$Ax = b,$$

$$A = \begin{bmatrix} 8 & -3 & 2 \\ 4 & 11 & -1 \\ 6 & 3 & 12 \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad b = \begin{bmatrix} 20 \\ 33 \\ 36 \end{bmatrix}.$$

$$x^* = (3, 2, 1)^{\mathrm{T}}.$$

$$\begin{cases} x_1 = \dfrac{1}{8}(3x_2 - 2x_3 + 20), \\ x_2 = \dfrac{1}{11}(-4x_1 + x_3 + 33), \\ x_3 = \dfrac{1}{12}(-6x_1 - 3x_2 + 36) \end{cases} \tag{8.1.3}$$

pdf-216

$$x = B_0 x + f,$$

$$B_0 = \begin{bmatrix} 0 & \dfrac{3}{8} & -\dfrac{2}{8} \\ -\dfrac{4}{11} & 0 & \dfrac{1}{11} \\ -\dfrac{6}{12} & -\dfrac{3}{12} & 0 \end{bmatrix} = I - D^{-1}A, \qquad f = \begin{bmatrix} \dfrac{20}{8} \\ \dfrac{33}{11} \\ \dfrac{36}{12} \end{bmatrix} = D^{-1}b.$$

$x^{(0)} = (0, 0, 0)^{\mathrm{T}}$

$$x^{(1)} = (x_1^{(1)}, x_2^{(1)}, x_3^{(1)})^{\mathrm{T}} = (2.5, 3, 3)^{\mathrm{T}},$$

$x^{(1)}$

$x^{(2)}$

$$x^{(0)} = \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \\ x_3^{(0)} \end{bmatrix}, \quad x^{(1)} = \begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \\ x_3^{(1)} \end{bmatrix}, \quad \cdots, \quad x^{(k)} = \begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \\ x_3^{(k)} \end{bmatrix}, \quad \cdots,$$

$$\begin{cases} x_1^{(k+1)} = (3x_2^{(k)} - 2x_3^{(k)} + 20)/8, \\ x_2^{(k+1)} = (-4x_1^{(k)} + x_3^{(k)} + 33)/11, \\ x_3^{(k+1)} = (-6x_1^{(k)} - 3x_2^{(k)} + 36)/12, \end{cases} \tag{8.1.4}$$

$$x^{(k+1)} = B_0 x^{(k)} + f,$$

k

$k = 0, 1, 2, \cdots$

$$x^{(10)} = (3.000\,032,\ 1.999\,838,\ 0.999\,881\,3)^{\mathrm{T}},$$

$$\|\varepsilon^{(10)}\|_\infty = 0.000\,187 \quad (\varepsilon^{(10)} = x^{(10)} - x^*).$$

$x^{(k)}$

x^*

$x = Bx + f$

$Ax = b$

$x^{(k)}$

x^*

$$\begin{cases} x_1 = 2x_2 + 5, \\ x_2 = 3x_1 + 5. \end{cases}$$

$x = Bx + f$

x^*

$$x^* = Bx^* + f. \tag{8.1.5}$$

$x^{(0)}$

$$\begin{cases} x^{(1)} = Bx^{(0)} + f, \\ x^{(2)} = Bx^{(1)} + f, \\ \quad\vdots \\ x^{(k+1)} = Bx^{(k)} + f, \end{cases} \tag{8.1.6}$$

$x^{(10)}$

1.999 838

0.000 187

$x^{(0)} = 0$

$(3.0000318141, 1.9998740186, 0.9998812605)^{\mathrm{T}}$

0.0001259814

0.000162

pdf-217

k

1°

$x = Bx + f$

B

k

2°

$\lim_{k\to\infty} x^{(k)}$

x^*

x^*

$\{x^{(k)}\}$

$$\varepsilon^{(k+1)} = x^{(k+1)} - x^*,$$

$$\varepsilon^{(k+1)} = B\varepsilon^{(k)} \quad (k = 0, 1, 2, \cdots),$$

$$\varepsilon^{(k)} = B\varepsilon^{(k-1)} = \cdots = B^k\varepsilon^{(0)}.$$

$\{x^{(k)}\}$

B

$\varepsilon^{(k)} \to 0$

$k \to \infty$

B

$B^k \to O$

$k \to \infty$

$$\sum_{j=1}^{n} a_{ij}x_j = b_i \quad (i = 1, 2, \cdots, n),$$

$$Ax = b, \tag{8.2.1}$$

A

$a_{ij} \neq 0$

$i = 1, 2, \cdots, n$

A

$A = D - L - U$

$$D = \begin{bmatrix} a_{11} & & & & \\ & a_{22} & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & a_{nn} \end{bmatrix}, \qquad L = -\begin{bmatrix} 0 & & & & \\ a_{21} & 0 & & & \\ a_{31} & a_{32} & 0 & & \\ \vdots & \vdots & \ddots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{n,n-1} & 0 \end{bmatrix},$$

$$U = -\begin{bmatrix} 0 & a_{12} & a_{13} & \cdots & a_{1n} \\ & 0 & a_{23} & \cdots & a_{2n} \\ & & \ddots & \ddots & \vdots \\ & & & 0 & a_{n-1,n} \\ & & & & 0 \end{bmatrix}.$$

i
$i = 1, 2, \cdots, n$
a_{ii}
$a_{ij} \neq 0$
$i = 1, 2, \cdots, n$
j
a_{ii}
$a_{ii} \neq 0$

pdf-218

$$x_i = \frac{1}{a_{ii}}\left(b_i - \sum_{\substack{j=1 \\ j\neq i}}^{n} a_{ij}x_j\right) \quad (i = 1, 2, \cdots, n), \tag{8.2.2}$$

$$x = B_0 x + f,$$

$$B_0 = I - D^{-1}A = D^{-1}(L+U), \qquad f = D^{-1}b.$$

$$\begin{cases} x^{(0)} = (x_1^{(0)}, x_2^{(0)}, \cdots, x_n^{(0)})^{\mathrm{T}} \quad \text{（初始向量）}, \\ x_i^{(k+1)} = \dfrac{1}{a_{ii}}\left(b_i - \displaystyle\sum_{\substack{i=1 \\ j\neq i}}^{n} a_{ij}x_j^{(k)}\right), \end{cases} \tag{8.2.3}$$

$x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \cdots, x_n^{(k)})^{\mathrm{T}}$
k
$x^{(k)}$
$x^{(k+1)}$
$k = 0, 1, 2, \cdots;\ i = 1, 2, \cdots, n$

$$\begin{cases} x^{(0)} \quad \text{（初始向量）}, \\ x^{(k+1)} = B_0 x^{(k)} + f, \end{cases} \tag{8.2.4}$$

B_0
$x^{(k)}$
$x^{(k+1)}$
$x^{(k)}$
$x^{(k+1)}$
i
$x_i^{(k+1)}$
$x_1^{(k+1)}, x_2^{(k+1)}, \cdots, x_{i-1}^{(k+1)}$
$k+1$
$x^{(k+1)}$
$x_j^{(k+1)}$

$$\begin{gathered} x^{(0)} = (x_1^{(0)}, x_2^{(0)}, \cdots, x_n^{(0)})^{\mathrm{T}} \quad \text{（初始向量）}, \\ x_i^{(k+1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij}x_j^{(k)}\right) \quad (k = 0, 1, 2, \cdots;\ i = 1, 2, \cdots, n), \end{gathered} \tag{8.2.5}$$

$$\begin{cases} x_i^{(k+1)} = x_i^{(k)} + \Delta x_i \quad (k = 0, 1, 2, \cdots;\ i = 1, 2, \cdots, n), \\ \Delta x_i = \dfrac{1}{a_{ii}}\left(b_i - \displaystyle\sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij}x_j^{(k)}\right). \end{cases} \tag{8.2.5'}$$

$x_1^{(k+1)}$
i
$x_j^{(k+1)}$
$j = 1, 2, \cdots, i-1$

$$Dx^{(k+1)} = b + Lx^{(k+1)} + Ux^{(k)}, \qquad (D-L)x^{(k+1)} = b + Ux^{(k)},$$

$(D-L)^{-1}$
$i = 1$

$j \neq i$
$j = 1$

pdf-219

$$x^{(k+1)} = (D-L)^{-1}Ux^{(k)} + (D-L)^{-1}b,$$

$$x^{(k+1)} = Gx^{(k)} + f, \tag{8.2.6}$$

$$G = (D-L)^{-1}U, \qquad f = (D-L)^{-1}b.$$

$x = Gx + f$
G
,
$x^{(0)} = 0$

$$\begin{cases} x_1^{(k+1)} = (20 + 3x_2^{(k)} - 2x_3^{(k)})/8, \\ x_2^{(k+1)} = (33 - 4x_1^{(k+1)} + x_3^{(k)})/11, \\ x_3^{(k+1)} = (36 - 6x_1^{(k+1)} - 3x_2^{(k+1)})/12. \end{cases}$$

$$x^{(5)} = (2.999\,843,\ 2.000\,072,\ 1.000\,061)^{\mathrm{T}}, \qquad \|\varepsilon^{(5)}\|_\infty = 0.001\,57.$$

$$\begin{cases} x_1 + 2x_2 - 2x_3 = 1, \\ x_1 + x_2 + x_3 = 1, \\ 2x_1 + 2x_2 + x_3 = 1 \end{cases}$$

$A_k = (a_{ij}^{(k)})_{n\times n}$
$k = 1, 2, \cdots$
$A = (a_{ij})_{n\times n}$

$$\lim_{k\to\infty} a_{ij}^{(k)} = a_{ij} \quad (i, j = 1, 2, \cdots, n)$$

$\{A_k\}$
A
$\lim\limits_{k\to\infty} A_k = A$

$$A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \quad A^2 = \begin{bmatrix} \lambda^2 & 2\lambda \\ 0 & \lambda^2 \end{bmatrix}, \quad \cdots, \quad A^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{bmatrix}, \quad \cdots,$$

$|\lambda| < 1$
$A^k \to \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
$k \to \infty$
0.001 57
$(3, 2, 1)^{\mathrm{T}}$
0.000157
0.0001576134

pdf-220

$\lim\limits_{k\to\infty} A_k = A$
$\|A_k - A\| \to 0$
$k \to \infty$
$\{x^{(k)}\}$
B

$$\varepsilon^{(k)} = x^{(k)} - x^* = B^k\varepsilon^{(0)} \to 0 \quad (k \to \infty).$$

$B = (b_{ij})_{n\times n}$
$B^k \to O$
$k \to \infty$
$\rho(B) < 1$
B
P

$$P^{-1}BP=\begin{bmatrix}J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_r\end{bmatrix}\equiv J,\quad \text{其中} J_i=\begin{bmatrix}\lambda_i & 1 & & & \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ & & & & \lambda_i\end{bmatrix}_{n_i\times n_i},$$

$\sum_{i=1}^{r} n_i = n$
$B=PJP^{-1}$

$$B^k=PJ^kP^{-1},$$

$$J^k=\begin{bmatrix}J_1^k & & & \\ & J_2^k & & \\ & & \ddots & \\ & & & J_r^k\end{bmatrix}\quad (k=1,2,\cdots).$$

$$B^k\to O\quad (k\to\infty)\iff J^k\to O\quad (k\to\infty),$$

$$J^k\to O\quad (k\to\infty)\iff J_i^k\to O\quad (k\to\infty)\quad (i=1,2,\cdots,r).$$

$$E_{tk}=\begin{bmatrix}\overbrace{0\ \cdots\ 0}^{k\text{个}} & 1 & 0 & \cdots & 0\\ \ddots & \ddots & \ddots & \ddots & \vdots\\ & \ddots & \ddots & \ddots & 0\\ & & \ddots & \ddots & 1\\ & & & \ddots & 0\\ & & & \ddots & \vdots\\ & & & & 0\end{bmatrix}_{t\times t}\quad (\text{其中} t=n_i),$$

E_{tk}
k
k
$1,0,\cdots,0$
k
1
$t\times t$
$t=n_i$
$(E_{t1})^k=E_{tk}$
$k\geqslant t$
$E_{tk}=O$

$$J_i=\begin{bmatrix}\lambda_i & & & & \\ & \lambda_i & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & \lambda_i\end{bmatrix}+\begin{bmatrix}0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1\\ & & & & 0\end{bmatrix}=\lambda_i I+E_{t1},$$

pdf-221

$$J_i^k=(\lambda_i I+E_{t1})^k=\sum_{j=0}^{k}\binom{k}{j}\lambda_i^{k-j}(E_{t1})^j=\sum_{j=0}^{k}\binom{k}{j}\lambda_i^{k-j}E_{tj}=\sum_{j=0}^{t-1}\binom{k}{j}\lambda_i^{k-j}E_{tj}$$

$$=\begin{bmatrix}\lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \cdots & \cdots & \binom{k}{t-1}\lambda_i^{k-(t-1)}\\ & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \cdots & \binom{k}{t-2}\lambda_i^{k-(t-2)}\\ & & \ddots & \ddots & \vdots\\ & & & \ddots & \binom{k}{1}\lambda_i^{k-1}\\ & & & & \lambda_i^k\end{bmatrix}_{t\times t}\quad (i=1,2,\cdots,r),$$

$$E_{t0}=I,\qquad \binom{k}{j}=\frac{k!}{j!(k-j)!}=\frac{k(k-1)\cdots(k-j+1)}{j!}.$$

$\lim_{k\to\infty} k^r c^k=0$
$0<c<1, r\geqslant 0$

$$\binom{k}{j}\lambda^{k-j} \to 0 \quad (k \to \infty) \iff |\lambda| < 1,$$

$J_i^k \to O$
$k \to \infty$
$|\lambda_i| < 1$
$i = 1, 2, \cdots, r$
$\rho(B) < 1$

$$x = Bx + f, \tag{8.3.1}$$

$x^{(0)}$
f
$x^{(k+1)} = Bx^{(k)} + f$
$\rho(B) < 1$
$\rho(B) < 1$
$Ax = f$
$A = I - B$

$$x^* = Bx^* + f. \tag{8.3.2}$$

$$\varepsilon^{(k)} = x^{(k)} - x^* = B^k \varepsilon^{(0)}, \qquad \varepsilon^{(0)} = x^{(0)} - x^*.$$

$\rho(B) < 1$
$B^k \to O$
$k \to \infty$
$x^{(0)}$
f
$\varepsilon^{(k)} \to 0$
$k \to \infty$
$x^{(k)} \to x^*$
$k \to \infty$
$x^{(0)}$
f
$\lim\limits_{k\to\infty} x^{(k)} = x^*$
$x^{(k+1)} = Bx^{(k)} + f$
x^*
$x^{(0)}$
f

$$\varepsilon^{(k)} = x^{(k)} - x^* = B^k \varepsilon^{(0)} \to 0 \quad (k \to \infty).$$

$$B^k \to O \quad (k \to \infty).$$

$\rho(B) < 1$

pdf-222

B_0

$$\det(\lambda I - B_0) = \lambda^3 + 0.034\,090\,909\lambda + 0.039\,772\,727 = 0,$$

$$\lambda_1 = -0.308\,2,$$
$$\lambda_2 = 0.154\,1 + \mathrm{i}0.324\,5,$$
$$\lambda_3 = 0.154\,1 - \mathrm{i}0.324\,5,$$
$$|\lambda_2| = |\lambda_3| = 0.359\,2 < 1, \qquad |\lambda_1| < 1,$$

$\rho(B_0) < 1$
$x^{(k+1)} = Bx^{(k)} + f$
$k = 0, 1, 2, \cdots$

$$B = \begin{pmatrix} 0 & 2 \\ 3 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 5 \\ 5 \end{pmatrix}.$$

$$\det(\lambda I - B) = \lambda^2 - 6 = 0,$$

$\lambda_{1,2} = \pm\sqrt{6}$
$\rho(B) > 1$
$\varepsilon^{(k)} = x^{(k)} - x^* = B^k \varepsilon^{(0)}$

B

n

$u_1, u_2, \cdots, u_n$

$\lambda_1, \lambda_2, \cdots, \lambda_n$

$\varepsilon^{(0)} = \sum\limits_{i=1}^{n} a_i u_i$

$$\varepsilon^{(k)} = B^k \varepsilon^{(0)} = \sum_{i=1}^{n} a_i B^k u_i = \sum_{i=1}^{n} a_i \lambda_i^k u_i.$$

$\rho(B) < 1$

$\lambda_i^k \to 0$

$i = 1, 2, \cdots, n;\ k \to \infty$

$\varepsilon^{(k)} \to 0$

$\rho(B)$

k

$$[\rho(B)]^k \leqslant 10^{-s}, \tag{8.3.3}$$

$$k \geqslant \frac{s \ln 10}{-\ln \rho(B)}.$$

$R(B) = -\ln \rho(B)$

$\rho(B) < 1$

$-\ln \rho(B)$

n

$\rho(B) \leqslant \|B\|_v$

$v = 1, 2, \infty$

F

$\|B\|_v$

$\rho(B)$

$x^{(k+1)} = Bx^{(k)} + f$

$x^{(0)}$

$\|B\|_v = q < 1$

1°

pdf-223

2°

$\|x^* - x^{(k)}\|_v \leqslant \dfrac{q}{1-q} \|x^{(k)} - x^{(k-1)}\|_v$

3°

$\|x^* - x^{(k)}\|_v \leqslant \dfrac{q^k}{1-q} \|x^{(1)} - x^{(0)}\|_v$

1°

2°

3°

$$x^* - x^{(k+1)} = B(x^* - x^{(k)}), \tag{8.3.4}$$

$$\|x^{(k+1)} - x^{(k)}\|_v \leqslant q \|x^{(k)} - x^{(k-1)}\|_v \quad (k = 1, 2, \cdots), \tag{8.3.5}$$

$$\|x^* - x^{(k+1)}\|_v \leqslant q \|x^* - x^{(k)}\|_v. \tag{8.3.6}$$

$$\begin{aligned} \|x^{(k+1)} - x^{(k)}\|_v &= \|x^* - x^{(k)} - (x^* - x^{(k+1)})\|_v \\ &\geqslant \|x^* - x^{(k)}\|_v - \|x^* - x^{(k+1)}\|_v \\ &\geqslant (1-q)\|x^* - x^{(k)}\|_v, \end{aligned}$$

$$\begin{aligned} \|x^* - x^{(k)}\|_v &\leqslant \frac{1}{1-q} \|x^{(k+1)} - x^{(k)}\|_v \\ &\leqslant \frac{q}{1-q} \|x^{(k)} - x^{(k-1)}\|_v \quad (k = 1, 2, \cdots). \end{aligned}$$

3°

B_0

∞

$$\|B_0\|_\infty = \max\left\{\frac{5}{8}, \frac{5}{11}, \frac{9}{12}\right\} = \frac{9}{12} < 1,$$

$x^{(k+1)} = Bx^{(k)} + f$

$k = 0, 1, 2, \cdots$

$$B = \begin{pmatrix} 0.9 & 0 \\ 0.3 & 0.8 \end{pmatrix}, \qquad f = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

$$\|B\|_\infty = 1.1, \qquad \|B\|_1 = 1.2, \qquad \|B\|_2 = 1.021, \qquad \|B\|_F = \sqrt{1.54}.$$

B

B

$\lambda_1 = 0.9, \lambda_2 = 0.8$

$\|B\| = q < 1$

B

$\|B\| < 1$

$\|x^{(k)} - x^{(k-1)}\| < \varepsilon_0$

ε_0

$\|x^* - x^{(k)}\| \leqslant \dfrac{q}{1-q}\varepsilon_0$

$\|x^{(k)} - x^{(k-1)}\| < \varepsilon_0$

$q \approx 1$

$\dfrac{q}{1-q}$

$\|x^{(k)} - x^{(k-1)}\|$

$\|\varepsilon^{(k)}\| = \|x^* - x^{(k)}\|$

3°

$\|\varepsilon^{(k)}\| < \varepsilon_0$

$\rho(G) < 1$

pdf-224

G

$A = (a_{ij})_{n\times n} \in \mathbb{R}^{n\times n}$

$\mathbb{C}^{n\times n}$

1°

A

$$|a_{ii}| > \sum_{\substack{j=1\\ j\neq i}}^{n} |a_{ij}| \quad (i = 1, 2, \cdots, n), \tag{8.3.7}$$

A

A

2°

$|a_{ii}| \geqslant \sum\limits_{\substack{j=1\\ j\neq i}}^{n} |a_{ij}|$

$i = 1, 2, \cdots, n$

A

$A = \begin{bmatrix} -4 & 1 & 0 & 0 \\ 1 & -4 & 1 & 0 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 1 & -4 \end{bmatrix}$

$B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

$A = (a_{ij})_n \in \mathbb{R}^{n\times n}$

$\mathbb{C}^{n\times n}$

$n \geqslant 2$

n

P

$$P^{\mathrm{T}}AP = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix} \tag{8.3.8}$$

A_{11}

r

A_{22}

$n - r$

$1 \leqslant r \leqslant n$

A

P

A

A

$Ax = b$

A

A

$Ax = b$

$P^{\mathrm{T}}AP(P^{\mathrm{T}}x) = P^{\mathrm{T}}b$

$$y = P^{\mathrm{T}}x = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \qquad P^{\mathrm{T}}b = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix},$$

y_1, d_1

r

$Ax = b$

$$\begin{cases} A_{11}y_1 + A_{12}y_2 = d_1, \\ A_{22}y_2 = d_2. \end{cases}$$

B

$P = I_{13}$

$1 \leqslant r \leqslant n$

$1 \leqslant r \leqslant n-1$

$r = n$

pdf-225

$$P^{\mathrm{T}}BP = \left[\begin{array}{c|cc} 2 & 1 & 0 \\ \hline 0 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right].$$

$A = (a_{ij})_n \in \mathbb{R}^{n\times n}$

$\mathbb{C}^{n\times n}$

A

A

$\det A = 0$

$Ax = 0$

$$x = (x_1, x_2, \cdots, x_n)^{\mathrm{T}}.$$

$|x_k| = \max\limits_{1\leqslant i\leqslant n} |x_i| \neq 0$

k

$$\sum_{j=1}^{n} a_{kj}x_j = 0,$$

$$|a_{kk}x_k| = \left|\sum_{\substack{j=1\\j\neq k}}^{n} a_{kj}x_j\right| \leqslant \sum_{\substack{j=1\\j\neq k}}^{n} |a_{kj}|\,|x_j| \leqslant |x_k|\sum_{\substack{j=1\\j\neq k}}^{n} |a_{kj}|,$$

$$|a_{kk}| \leqslant \sum_{\substack{j=1\\j\neq k}}^{n} |a_{kj}|, \tag{8.3.9}$$

$\det A \neq 0$

A

$A \in \mathbb{R}^{n\times n}$

$x^{(0)}$

A

$a_{ii} \neq 0$

$i = 1, 2, \cdots, n$

$$G = (D-L)^{-1}U.$$

$$\begin{aligned} \det(\lambda I - G) &= \det(\lambda I - (D-L)^{-1}U) \\ &= \det(D-L)^{-1} \cdot \det(\lambda(D-L) - U) = 0, \end{aligned}$$

$$\det(\lambda(D-L) - U) = 0.$$

$$G = \lambda(D-L) - U = \begin{bmatrix} \lambda a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ \lambda a_{21} & \lambda a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \lambda a_{32} & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & a_{n-1,n} \\ \lambda a_{n1} & \lambda a_{n2} & \cdots & \lambda a_{n,n-1} & \lambda a_{nn} \end{bmatrix} = (c_{ij})_{n\times n}.$$

$|\lambda| \geqslant 1$

$\det G \neq 0$

$\det G = 0$

$|\lambda| < 1$

A

G
A
G
$G = \lambda(D-L)-U$
$\det G$

pdf-226

$|\lambda| \geqslant 1$

$$|c_{ii}| = |\lambda|\,|a_{ii}| \geqslant \sum_{j=1}^{i-1} |\lambda a_{ij}| + \sum_{j=i+1}^{n} |a_{ij}| = \sum_{j\neq i} |c_{ij}|,$$

$|\lambda| \geqslant 1$
G
$|\lambda| \geqslant 1$
$\det G \neq 0$
$\rho(G) < 1$

$$Ax = b, \tag{8.4.1}$$

$A \in \mathbb{R}^{n\times n}$
$a_{ii} \neq 0$
$i = 1, 2, \cdots, n$
A

$$A = D - L - U. \tag{8.4.2}$$

k
$x^{(k)}$
$k+1$
$x^{(k+1)}$
$x_j^{(k+1)}$
$j = 1, 2, \cdots, i-1$
$x_i^{(k+1)}$

$$\tilde{x}_i^{(k+1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)}\right) \quad (i = 1, 2, \cdots, n), \tag{8.4.3}$$

$x_i^{(k+1)}$
$x_i^{(k)}$
$\tilde{x}_i^{(k+1)}$

$$x_i^{(k+1)} = (1-\omega)x_i^{(k)} + \omega\tilde{x}_i^{(k+1)} = x_i^{(k)} + \omega(\tilde{x}_i^{(k+1)} - x_i^{(k)}). \tag{8.4.4}$$

$Ax = b$

$$\begin{cases} x_i^{(k+1)} = x_i^{(k)} + \dfrac{\omega}{a_{ii}}\left(b_i - \displaystyle\sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij} x_j^{(k)}\right), \\ x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \cdots, x_n^{(k)})^{\mathrm{T}} \quad (k = 0, 1, \cdots;\ i = 1, 2, \cdots, n). \end{cases} \tag{8.4.5}$$

ω

$$\begin{cases} x_i^{(k+1)} = x_i^{(k)} + \Delta x_i \quad (k = 0, 1, \cdots;\ i = 1, 2, \cdots, n), \\ \Delta x_i = \dfrac{\omega}{a_{ii}}\left(b_i - \displaystyle\sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij} x_j^{(k)}\right). \end{cases} \tag{8.4.6}$$

$\omega = 1$

pdf-227

$|p_0| = \max_i |\Delta x_i| = \max_{1\leqslant i\leqslant n} |x_i^{(k+1)} - x_i^{(k)}| < \varepsilon$
$\omega < 1$
$\omega > 1$

$$\begin{bmatrix} -4 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 \\ 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

$x^* = (-1, -1, -1, -1)^{\mathrm{T}}$
$x^{(0)} = 0$

$$\begin{cases} x_1^{(k+1)} = x_1^{(k)} - \omega(1 + 4x_1^{(k)} - x_2^{(k)} - x_3^{(k)} - x_4^{(k)})/4, \\ x_2^{(k+1)} = x_2^{(k)} - \omega(1 - x_1^{(k+1)} + 4x_2^{(k)} - x_3^{(k)} - x_4^{(k)})/4, \\ x_3^{(k+1)} = x_3^{(k)} - \omega(1 - x_1^{(k+1)} - x_2^{(k+1)} + 4x_3^{(k)} - x_4^{(k)})/4, \\ x_4^{(k+1)} = x_4^{(k)} - \omega(1 - x_1^{(k+1)} - x_2^{(k+1)} - x_3^{(k+1)} + 4x_4^{(k)})/4. \end{cases}$$

$\omega = 1.3$

$$x^{(11)} = (-0.999\,996\,46,\ -1.000\,003\,10,\ -0.999\,999\,53,\ -0.999\,999\,12)^{\mathrm{T}},$$

$$\|\varepsilon^{(11)}\|_2 \leqslant 0.46 \times 10^{-5}.$$

ω

$\omega = 1.3$

ω

$\|x^{(k)} - x^*\|_2 < 10^{-5}$

$\underline{1.3}$

$$a_{ii}x_i^{(k+1)} = (1-\omega)a_{ii}x_i^{(k)} + \omega\left(b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij}x_j^{(k)}\right) \quad (i = 1, 2, \cdots, n). \tag{8.4.7}$$

$A = D - L - U$

$$Dx^{(k+1)} = \omega(b + Lx^{(k+1)} + Ux^{(k)}) + (1-\omega)Dx^{(k)},$$

$$(D - \omega L)x^{(k+1)} = ((1-\omega)D + \omega U)x^{(k)} + \omega b.$$

ω

$D - \omega L$

$a_{ii} \neq 0;\ i = 1, 2, \cdots, n$

$$x^{(k+1)} = (D - \omega L)^{-1}[(1-\omega)D + \omega U]x^{(k)} + \omega(D - \omega L)^{-1}b.$$

$a_{ii} \neq 0;\ i = 1, 2, \cdots, n$

$$x^{(k+1)} = L_\omega x^{(k)} + f. \tag{8.4.8}$$

$\|x^{(11)} - x^*\|_2 \approx 4.8100832 \times 10^{-6}$

0.46×10^{-5}

$\omega = 13/10$

$(-0.9999966722, -1.0000028727, -0.9999995354, -0.9999991925)^{\mathrm{T}}$

4.4938646×10^{-6}

pdf-228

$$L_\omega = (D - \omega L)^{-1}[(1-\omega)D + \omega U], \qquad f = \omega(D - \omega L)^{-1}b.$$

L_ω

$x = L_\omega x + f$

$a_{ii} \neq 0$

$i = 1, 2, \cdots, n$

$$\rho(L_\omega) < 1.$$

ω

ω

$\rho(L_\omega) = \min_{\omega}$

$a_{ii} \neq 0;\ i = 1, 2, \cdots, n$

ω

$a_{ii} \neq 0;\ i = 1, 2, \cdots, n$

$$0 < \omega < 2.$$

$\rho(L_\omega) < 1$

L_ω

$\lambda_1, \lambda_2, \cdots, \lambda_n$

$$|\det L_\omega| = |\lambda_1 \lambda_2 \cdots \lambda_n| \leqslant (\rho(L_\omega))^n, \qquad |\det L_\omega|^{1/n} \leqslant \rho(L_\omega) < 1.$$

$$\det L_\omega = \det((D - \omega L)^{-1}) \cdot \det((1-\omega)D + \omega U) = (1-\omega)^n,$$

$$|1-\omega| < 1.$$

$a_{ii} \neq 0;\ i = 1, 2, \cdots, n$

ω

$(0, 2)$

A

ω

$0 < \omega < 2$

A

$0 < \omega < 2$

$|\lambda| < 1$

λ

L_ω

y

λ

L_ω

$$L_\omega y = \lambda y, \qquad y = (y_1, y_2, \cdots, y_n)^{\mathrm{T}} \neq 0,$$

$$(D - \omega L)^{-1}[(1-\omega)D + \omega U]y = \lambda y,$$

$$[(1-\omega)D + \omega U]y = \lambda(D - \omega L)y.$$

λ

$$(((1-\omega)D + \omega U)y, y) = \lambda((D - \omega L)y, y),$$

$$\lambda = \frac{(Dy, y) - \omega(Dy, y) + \omega(Uy, y)}{(Dy, y) - \omega(Ly, y)},$$

pdf-229

$$(Dy, y) = \sum_{i=1}^{n} a_{ii}|y_i|^2 \equiv \sigma > 0, \tag{8.4.9}$$

$-(Ly, y) = \alpha + \mathrm{i}\beta$

$A = A^{\mathrm{T}}$

$U = L^{\mathrm{T}}$

$$-(Uy, y) = -(y, Ly) = -\overline{(Ly, y)} = \alpha - \mathrm{i}\beta,$$

$$0 < (Ay, y) = ((D - L - U)y, y) = \sigma + 2\alpha, \tag{8.4.10}$$

$$\lambda = \frac{(\sigma - \omega\sigma - \alpha\omega) + \mathrm{i}\omega\beta}{(\sigma + \alpha\omega) + \mathrm{i}\omega\beta},$$

$$|\lambda|^2 = \frac{(\sigma - \omega\sigma - \alpha\omega)^2 + \omega^2\beta^2}{(\sigma + \alpha\omega)^2 + \omega^2\beta^2}.$$

$0 < \omega < 2$

$$(\sigma - \omega\sigma - \alpha\omega)^2 - (\sigma + \alpha\omega)^2 = \omega\sigma(\sigma + 2\sigma)(\omega - 2) < 0,$$

L_ω

$|\lambda| < 1$

$0 < \omega < 2$

$$(\sigma + 2\omega)^2 + \omega^2\beta^2 \neq 0.$$

$Ax = b$

$$\omega_{\mathrm{opt}} = \frac{2}{1 + \sqrt{1 - \rho^2(B_0)}},$$

$\rho(B_0)$

B_0

$\rho(B_0)$

A

x

$x^{(k)}$

$$|p_0| = \max_{1\leqslant i\leqslant n} |\Delta x_i| = \max_{1\leqslant i\leqslant n} |x_i^{(k+1)} - x_i^{(k)}| < \varepsilon \quad \text{（精度要求）}$$

k

$k \leftarrow 0$

$x_i \leftarrow 0$

$i = 1, 2, \cdots, n$

$k \leftarrow k+1$

$p_0 \leftarrow 0$

$i = 1, 2, \cdots, n$

$p \leftarrow \Delta x_i = \omega \left(b_i - \sum_{j=1}^{i-1} a_{ij}x_j - \sum_{j=i+1}^{n} a_{ij}x_j \right) / a_{ii}$

$|p| > |p_0|$

$p_0 \leftarrow p$

$x_i \leftarrow x_i + p$

p_0

$\sigma + 2\sigma$

$\sigma + 2\alpha$

$(\sigma + 2\omega)^2 + \omega^2\beta^2 \neq 0$

$|\lambda|^2$

$(\sigma + \alpha\omega)^2 + \omega^2\beta^2$

$j = i+1$

$x_i \leftarrow x_i + p$

$p = \Delta x_i$

$j = i$

$a_{ii}x_i$

pdf-230

$|p_0| > \varepsilon$

x, k

$Ax = b$

A

A

$$\begin{cases} 5x_1 + 2x_2 + x_3 = -12, \\ -x_1 + 4x_2 + 2x_3 = 20, \\ 2x_1 - 3x_2 + 10x_3 = 3. \end{cases}$$

$\|x^{(k+1)} - x^{(k)}\|_\infty < 10^{-4}$

$A = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}$

$\|A\|_1 = \|A\|_\infty > 1$

$I + A + A^2 + \cdots + A^k + \cdots$

A

$I, A, \frac{1}{2}A^2, \frac{1}{3!}A^3, \frac{1}{4!}A^4, \cdots$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases} \quad (a_{11} \cdot a_{22} \neq 0).$$

$$\begin{cases} x_1^{(k)} = \dfrac{1}{a_{11}}(b_1 - a_{12}x_2^{(k-1)}), \\ x_2^{(k)} = \dfrac{1}{a_{22}}(b_2 - a_{21}x_1^{(k-1)}) \end{cases} \quad (k = 1, 2, \cdots).$$

$\{x^{(k)}\}$

$r = \left| \frac{a_{12}a_{21}}{a_{11}a_{22}} \right| < 1$

pdf-231

$$\begin{cases} x_1 + 0.4x_2 + 0.4x_3 = 1, \\ 0.4x_1 + x_2 + 0.8x_2 = 2, \\ 0.4x_1 + 0.8x_2 + x_3 = 3; \end{cases}$$

$$\begin{cases} x_1 + 2x_2 - 2x_3 = 1, \\ x_1 + x_2 + x_3 = 1, \\ 2x_1 + 2x_2 + x_3 = 1. \end{cases}$$

$\lim_{k\to\infty} A_k = A$

x

$\lim_{k\to\infty} A_k x = Ax$
$Ax = b$
A

$$\begin{cases} x_1 - \frac{1}{4}x_3 - \frac{1}{4}x_4 = \frac{1}{2}, \\ x_2 - \frac{1}{4}x_3 - \frac{1}{4}x_4 = \frac{1}{2}, \\ -\frac{1}{4}x_1 - \frac{1}{4}x_2 + x_3 = \frac{1}{2}, \\ -\frac{1}{4}x_1 - \frac{1}{4}x_2 + x_4 = \frac{1}{2}. \end{cases}$$

B_0
B_0
$\omega = 1.03, \omega = 1, \omega = 1.1$

$$\begin{cases} 4x_1 - x_2 = 1, \\ -x_1 + 4x_2 - x_3 = 4, \\ -x_2 + 4x_3 = -3. \end{cases}$$

$x^* = \left(\frac{1}{2}, 1, -\frac{1}{2}\right)^{\mathrm{T}}$
$\|x^* - x^{(k)}\|_\infty < 5 \times 10^{-6}$
ω
$\omega = 0.9$

$$\begin{cases} 5x_1 + 2x_2 + x_3 = -12, \\ -x_1 + 4x_2 + 2x_3 = 20, \\ 2x_1 - 3x_2 + 10x_3 = 3, \end{cases}$$

$\|x^{(k+1)} - x^{(k)}\|_\infty < 10^{-4}$
$Ax = b$
A
$x^{(k+1)} = x^{(k)} + \omega(b - Ax^{(k)})$
$k = 0, 1, 2, \cdots$
$0 < \omega < \frac{2}{\beta}$
$0 < \alpha \leqslant \lambda(A) \leqslant \beta$
$Ax = b$
$x_i^{(k+1)}$
$x^{(k+1)}$
i

$$r_i^{(k+1)} = b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij} x_i^{(k)}.$$

$x_i^{(k+1)} = x_i^{(k)} + \frac{r_i^{(k+1)}}{a_{ii}}$
$\varepsilon^{(k)} = x^{(k)} - \lambda^*$
x^*
$\varepsilon_i^{(k+1)} = \varepsilon_i^{(k)} - \frac{r_i^{(k+1)}}{a_{ii}}$
$0.4x_1 + x_2 + 0.8x_2 = 2$
$0.8x_3$
B_0
$x_i^{(k)}$
$x_j^{(k)}$
$x^{(k)} - \lambda^*$
x^*
$\varepsilon^{(k)} = x^{(k)} - x^*$

pdf-232

$$r_i^{(k+1)} = \sum_{j=1}^{i-1} a_{ij} \varepsilon_j^{(k+1)} + \sum_{j=i}^{n} a_{ij} \varepsilon_j^{(k)};$$

A
$Q(\varepsilon^{(k)}) = (A\varepsilon^{(k)}, \varepsilon^{(k)})$

$$Q(\varepsilon^{(k+1)}) - Q(\varepsilon^{(k)}) = -\sum_{j=1}^{n} \frac{(r_j^{(k+1)})^2}{a_{jj}}.$$

A
$x^{(0)}$

A

A

B

n

A

$Az_1 + Bz_2 = b_1$

$Bz_1 + Az_2 = b_2$

$z_1, z_2, b_1, b_2 \in \mathbb{R}^n$

$$Az_1^{(m+1)} = b_1 - Bz_2^{(m)}, \qquad Az_2^{(m+1)} = b_2 - Bz_1^{(m)} \quad (m \geqslant 0);$$

$$Az_1^{(m+1)} = b_1 - Bz_2^{(m)}, \qquad Az_2^{(m+1)} = b_2 - Bz_1^{(m+1)} \quad (m \geqslant 0).$$

$A = \begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$

$-\dfrac{1}{2} < a < 1$

$-\dfrac{1}{2} < a < \dfrac{1}{2}$

$A = \begin{bmatrix} 5 & 1 & 2 & 3 \\ 0 & 2 & 0 & 4 \\ 3 & -1 & 2 & -1 \\ 0 & 3 & 0 & 7 \end{bmatrix}$

A

$x^{(k+1)} = Cx^{(k)} + g$

$C \in \mathbb{R}^{n \times n}$

$k = 0, 1, 2, \cdots$

C

$\lambda_i(C) = 0$

$i = 1, 2, \cdots, n$

n

A

$0 < \omega \leqslant 1$

$Ax = b$

$Ax = b$

A

$A^{\mathrm{T}}A$

$\mathrm{cond}(A^{\mathrm{T}}A)_2 = (\mathrm{cond}(A)_2)^2$

A

$\varepsilon^{(k)} = x^{(k)} - \lambda^*$

$x^* - x^{(k)}$